

Program : Diploma in Architecture	
Course Code : 6188	Course Title: Advance Computer Application Lab
Semester : 6	Credits: 2.5
Course Category: Program Core	
Periods per week: 4 (L:0, T:1, P:3)	Periods per semester: 60

Course Objectives:

- To acquire knowledge of advanced 3D modelling concepts and apply this knowledge to understand the architectural design work flow process in the industry.
- To create 3D BIM modeling with Parametric Software- Revit/ ArchiCAD and work with BIM content.

Course Pre-requisites:

Topic	Course code	Course name	Semester
Drafting knowledge		Architectural drawing - I, II & III	3, 4, 5
Auto CAD 2D drafting		Working drawing – II, III (CAD Lab)	4, 5

Course Outcomes:

On completion of the course, student will be able to:

Con	Description	Duration (Hours)	Cognitive Level
CO1	Introduce Architectural 3D modeling using SketchUp & Revit Architecture/ Archi CAD	18	Applying
CO2	Introduce rendering process and enhance digital images using Adobe Photoshop, Paint Shop Pro & other graphic software	15	Applying
CO3	Introduce BIM using ArchiCAD /Revit software	12	Applying
CO4	Develop BIM content:	12	Applying
	Lab Test	3	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1				3			3
CO2				3			3
CO3				3			3
CO4				3			3

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcome	Description	Duration (Hours)	Cognitive Level
CO1	Introduce Architectural 3D modeling using Sketch Up & Revit Architecture/Archi CAD		
M1.01	Familiarize 3D modeling with SketchUp & Revit Architecture/ Archi CAD	9	Understanding
M1.02	Develop Architectural 3D modeling with SketchUp & Revit Architecture/ Archi CAD	9	Applying
CO2	To introduce rendering process and enhance digital images using Adobe Photoshop, Paint Shop Pro & other graphic software		
M2.01	Familiarize rendering techniques using SketchUp & Revit Architecture/ Archi CAD	4	Understanding
M2.02	Familiarize enhancement techniques of digital images using Adobe Photoshop, Paint Shop Pro & other graphic softwares	4	Understanding
M2.03	Demonstrate rendering process using SketchUp & Revit Architecture/ Archi CAD	4	Applying
M2.04	Demonstrate enhancing techniques of digital images from CAD software using Adobe Photoshop, Paint Shop Pro & other graphic softwares	3	Applying
	Lab Test I	1.5	
CO3	To introduce BIM using ArchiCAD /Revit software		
M3.01	Introduce ArchiCAD/Revit software	4	Understanding
M3.02	Discuss BIM tools	4	Understanding
M3.03	Familiarize families, Massing Tools,	4	Nderstanding

	Annotations and dimensions		
CO4	To develop BIM content		
M4.01	Develop technical drawings	6	Applying
M4.02	Generate specifications and estimates	6	Applying
	Lab Test II	1.5	

Text / Reference:

T/R	Book Title/Author
R1	Mastering AutoCAD 2016 and AutoCAD LT 2016: by George Omura& Brian C. Benton (Jul 2014)
R2	Mastering Autodesk Revit Architecture 2016: – by James Vandezande, Eddy Krygiel & Phil Read, Sybex; First edition
R3	Rendering in SketchUp: From Modeling to Presentation for Architecture, Landscape Architecture and Interior Design by Daniel Tal, John Wiley & Sons
R4	The SketchUp Workflow for Architecture: Modeling Buildings, Visualizing Design, and Creating Construction Documents with SketchUp Pro and LayOut – by Michael Brightman, John Wiley & Sons.
R5	BIM tools and design intent. Limitations and opportunities", Cinti Luciani, S. Garagnani, R. Mingucci

Online Resources:

Sl. No	Website Link
1	https://www.sketchupschool.com/sketchup-guide www.sgoinstitute.in › Downloads › Civil Downloads › Manual ConcreteTech
2	https://static.sdcpublishations.com/pdfsampl/978-1-63057-111-5-2.pdf
3	https://archicadtutorials.com/
4	http://bimboom.blogspot.com/2007/02/revit-history_11.html
5	https://www.autodesk.in/products/revit/architecture